

TopP.pyx

```
1 import os
2 import openai
3 import secret
4 openai.api_key=secret.api_key
5
6 # WRITE YOUR CODE HERE
```

Top P

top_p, an alternative to sampling with temperature, is also referred to as nucleus sampling. Generally, it is not recommended to alter both the temperature and the `top_p`. `top_p` controls how many random results should be considered for completion as per the temperature. If we set so 0.1 means only the tokens comprising the top 10 probability mass are considered.

Remember how when using the temperature of 1 we had different answers show up. Let's try setting the `top_p=0.1`, meaning only top 10 of probable answers.

```
prompts = "Write a tagline for an ice cream shop"

response = openai.Completion.create(model="text-davinci-002",
                                   prompt=prompts,
                                   n=5,
                                   top_p=0.1)

for i in (response["choices"]):
    print(i["text"].strip())
```

TRY IT!

We can use our probabilities tool to see the probability associated with each token. The Probability tool can be accessed in the OpenAI playground. Make sure we toggle the show probability tab.

Stop sequences

Enter sequence and press Tab

Top P0.56

Frequency penalty0

Presence penalty0

Best of1

Inject start text

☒

Inject restart text

☒

Show probabilities

Full spectrum



Here we can see that the response generated 5 times was the same. That is because it is taking the top 10 most probable responses which is the same. What happens when we switch `top_p=0.8`.

```
response = openai.Completion.create(model="text-davinci-002",
                                   prompt=prompts,
                                   n=5,
                                   top_p=0.8)

for i in (response["choices"]):
    print(i["text"].strip())
```

TRY IT!

We start to see more variation as it has a bigger pool of responses to choose from.

With your understanding of both `temperature` and `top_p`, What would happen if we set `temperature = 1` and `top_p = 0.1`? Feel free to try it in your editors with an `n=5`.

- ☒ Higher temperatures will generate a more diverse response. However `top_p` will narrow the scope of possible answers to the top 10%. Randomly choosing between the top 10% will give us more or less the same completion
- ☐ Higher temperatures will generate a more diverse response. We will get multiple different answers
- ☐ Higher temperatures will generate a more diverse response. `top_p` will not be considered because you cannot use `top_p` and temperature at the same time.
- ☐ We will get more or less the same answers because at higher temperatures, will generate more stable responses.

The function for adding randomness (temperature) was set to max. However, the pool of choices was narrowed by `top_p` which makes it harder to generate appropriate random answers.

Check it!

Next